

Ratio, proportion and percentages - Set 2 - Solutions

Specification M3.1-M3.11.

Answer key

Question	Answer	Topic
1	D	Ratio, proportion and percentages
2	F	Ratio, proportion and percentages
3	E	Ratio, proportion and percentages

Worked solutions

Question 1 RAT-09 · Ratio, proportion and percentages · spec M3.9

p is directly proportional to $\sqrt{q}$, and q is inversely proportional to r . Which of the following is true?

Answer: D

$$p = a\sqrt{q} \text{ and } q = \frac{b}{r}, \text{ so } p = a\sqrt{\frac{b}{r}} = \frac{a\sqrt{b}}{\sqrt{r}}.$$

Full method - how and why

Chaining two proportionality statements. Write each with a constant and substitute.

- 1 Write them algebraically: $p = a\sqrt{q}$ and $q = \frac{b}{r}$ for constants $a, b > 0$.
why: 'Directly proportional' means a constant multiple; 'inversely' means a constant over.
- 2 Substitute: $p = a\sqrt{\frac{b}{r}} = \frac{a\sqrt{b}}{\sqrt{r}}$.
why: The square root distributes over the quotient.
- 3 Since $a\sqrt{b}$ is a constant, $p = \frac{\text{constant}}{\sqrt{r}}$: p is inversely proportional to $\sqrt{r}$.
why: Collect all the constants into one; only the r -dependence matters.

Where it goes wrong

- Forgetting the square root and concluding $p \propto \frac{1}{r}$ - an offered option.
- Losing the 'inversely' and choosing $p \propto \sqrt{r}$.

Takeaway Chain proportionalities by substituting; the accumulated constants never affect the final relationship.

Question 2 RAT-11 · Ratio, proportion and percentages · spec M3.1

On a map with scale 1 : 25 000, a lake has area 8 cm^2 . What is the actual area of the lake in km^2 ?

Answer: F

$$1 \text{ cm represents } 25\,000 \text{ cm} = 0.25 \text{ km, so } 1 \text{ cm}^2 \text{ represents } 0.0625 \text{ km}^2. 8 \times 0.0625 = 0.5 \text{ km}^2.$$

Full method - how and why

A map scale applies to **lengths**. For areas the scale factor is squared - and the unit change must be handled too.

- 1 Convert the length scale: 1 cm on the map is 25 000 cm in reality = 250 m = 0.25 km.
why: $25\,000 \text{ cm} = 250 \text{ m}$ (divide by 100), then $= 0.25 \text{ km}$ (divide by 1000).
- 2 Square it for areas: 1 cm^2 represents $(0.25)^2 = 0.0625 \text{ km}^2$.
why: Area scales with the square of the length factor.
- 3 Multiply: $8 \times 0.0625 = 0.5 \text{ km}^2$.
why: $8 \times \frac{1}{16} = \frac{1}{2}$.

Where it goes wrong

- Using the length factor directly: $8 \times 0.25 = 2 \text{ km}^2$ - an offered option.
- Mishandling the cm-to-km conversion by a factor of 10, giving 0.05 or 5.

Takeaway Map areas use the **square** of the linear scale; convert units before squaring.

Question 3 RAT-12 · Ratio, proportion and percentages · spec M3.6

In a year group the ratio of boys to girls is 3 : 4. Two boys leave and two girls join, and the ratio becomes 2 : 3. How many students are in the year group?

Answer: E

With $3k$ boys and $4k$ girls: $\frac{3k-2}{4k+2} = \frac{2}{3} \Rightarrow 9k-6 = 8k+4 \Rightarrow k = 10$. Total $7k = 70$.

Full method - how and why

A ratio before and after a change. Introduce a multiplier k so both quantities are expressed with one unknown.

- 1 Let the numbers be $3k$ boys and $4k$ girls.

why: The ratio 3 : 4 fixes only the proportion, so a scale factor k is needed.

- 2 After the change: $3k - 2$ boys and $4k + 2$ girls, with ratio 2 : 3.

why: Two boys leave, two girls join.

- 3 Write the equation: $\frac{3k-2}{4k+2} = \frac{2}{3}$, so $3(3k-2) = 2(4k+2)$.

why: Cross-multiply.

- 4 Solve: $9k - 6 = 8k + 4 \Rightarrow k = 10$, so the total is $7k = 70$.

why: Check: 30 boys and 40 girls to start; after the change, $28 : 42 = 2 : 3$ ✓.

Where it goes wrong

- Forgetting to answer the total and giving k or one of the groups.
- Writing the new ratio upside down.

Takeaway Introduce k for a ratio, apply the change, cross-multiply, then convert back to the quantity asked for.